



Smart Pantry & Meal Finder

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The Problem

People have food at home, but still ask: “What can I cook now?”

Forgotten ingredients

Ingredients stay in the pantry or fridge until they expire, because users forget what they already have.

Unnecessary shopping

Users buy more groceries even when they already have enough to cook a meal.

Hard meal decisions

Choosing a recipe becomes harder when quantities, missing items, and available options are not clear.

The Solution

Loot turns pantry ingredients into clear meal options.

Track ingredients

Users store ingredients, units, quantities, and low-stock thresholds.

Find recipes

Loot shows what can be cooked, what is almost possible, and exactly what is missing.

Cook smarter

After cooking, quantities are deducted and cooking history is saved for repeat meals.

لوت helps users reduce waste, save time, and make better cooking decisions.

New Technology Used

Spring Mail

Used for system emails such as welcome messages, password reset codes, and low-stock ingredient lists. It adds a communication layer between Loot and the user.

OpenAI Integration

Used in the AI service to understand images, generate recipes, recommend meals, suggest substitutes, and rescue leftovers based on pantry data.

Both technologies support the core idea: smarter decisions + better communication.

How Loot Works



Login



POST

`/api/v1/user/login`

Purpose

Authenticates the user and creates the session used by the rest of the app.

Flow

Receives email and password →
validates credentials → stores userId in
HttpSession.

Output

Success: logged-in user ID.
Errors: email not found, wrong
password.

Missing Ingredients

GET

`/api/v1/user/missing/{listType}/{recipeId}`

Purpose

Shows exactly what the user is missing to cook a selected recipe.

Flow

Checks recipe type: system or user → compares required quantities with pantry quantities.

Output

Returns a missing list with required, available, and missing quantities.

System Recipes by Category

GET

`/api/v1/user/system/{category}`

Purpose

Displays system recipes inside one category such as breakfast, lunch, dinner, or snack.

Flow

Validates category → fetches matching system recipes → returns recipe DTO list.

Output

Returns recipe name, description, instructions, and ingredient list.

Possible System Recipes

GET

`/api/v1/user/possible/system/{category}`

Purpose

Finds system recipes the user can cook now using current pantry stock.

Flow

Validates category → checks every recipe ingredient against pantry quantities.

Output

Returns only recipes where all ingredients are available in enough quantity.

Almost Possible History Recipes

GET

`/api/v1/user/almost/history/{category}`

Purpose

Looks at previous cooked recipes and shows meals that are close to being possible again.

Flow

Filters history by category → compares each historical recipe with pantry quantities.

Output

Returns almost possible recipes with a missing ingredients list.

AI Image → Ingredient

POST

`/api/v1/ai/image/to/ingredient`

Purpose

Identifies an ingredient from an uploaded image.

Flow

User uploads JPEG, PNG, or WEBP → image is processed → AI predicts the ingredient name and unit.

Output

Returns ImageToIngredientDTO, or a clear message if the ingredient cannot be identified.

AI Image → Ingredient Add

POST

`/api/v1/ai/image/to/ingredient/add`

Purpose

Adds the AI-detected ingredient to the pantry.

Flow

Receives the detected ingredient DTO → checks user, ingredient, unit, and pantry state.

Output

Can add a new ingredient, add to pantry, or update pantry quantity.

AI Ingredient Substitute

POST

`/api/v1/ai/ingredient/substitute/{recipeId}/{listType}/{ingredientId}`

Purpose

Suggests a substitute for a missing recipe ingredient based on the pantry.

Flow

Checks recipe and ingredient → sends pantry context to AI → validates the suggested substitute.

Output

Returns a substitute suggestion, or says no suitable substitute is available.

AI Recipe Recommendation

POST

`/api/v1/ai/recipe/recommendation`

Purpose

Recommends recipes based on a user request and pantry context.

Flow

User writes a request → pantry is converted to AI context → AI returns matched ideas.

Output

Returns a list of AIRecipeDTO recommendations.

AI Leftover Rescue

POST

`/api/v1/ai/leftover/rescue`

Purpose

Turns leftovers into new recipe ideas.

Flow

User enters leftover ingredients → AI generates meals that reuse them with pantry support.

Output

Returns recipe ideas designed around the provided leftovers.

AI Recipe Generator



POST

`/api/v1/ai/recipe/generator`

Purpose

Generates a new full recipe from the user's request.

Flow

User describes what they want → AI creates name, category, instructions, and ingredients.

Output

Returns a GeneratedRecipeDTO ready for review.

AI Recipe Generator Add



POST

`/api/v1/ai/recipe/generator/add`

Purpose

Saves an AI-generated recipe as a user recipe.

Flow

Receives GeneratedRecipeDTO →
validates recipe and ingredients →
stores it under the logged-in user.

Output

Success: generated recipe added.
Errors: duplicate name or invalid data.

Core Features

The feature slide can be adjusted after you choose the final focus.

Pantry Intelligence

- Ingredient tracking
- Low-stock detection
- Quantity-based matching

Recipe Discovery

- Possible recipes
- Almost possible recipes
- Missing ingredient details

AI Assistance

- Image recognition
- Substitutes
- Recommendations
- Recipe generation



Thank You

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